

Total No. of Questions : 10]

SEAT No. :

P194

[Total No. of Pages : 2

[5871]-716

B.E. (Computer Engineering)

EMBEDDED AND REAL TIME OPERATING SYSTEM

(2015 Pattern) (Elective - III) (Semester - II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What is embedded system? List and explain different challenges of embedded system. [5]
b) What are the different types of processor technologies used in embedded system design? [5]

OR

- Q2)** a) List and Explain software tools used for designing of an embedded system. [5]
b) Describe different forms of memories in the embedded system. [5]

- Q3)** a) Enlist various Internet enabled system protocols and explain one with its features. [5]
b) Explain least slack time first scheduling and latest release time scheduling in real time systems. [5]

OR

- Q4)** a) Explain Serial protocol RS-232C in detail with neat diagram. [5]
b) Enlist various Internet enabled embedded system protocols and explain any one with its features and diagram. [5]

- Q5)** a) What is priority inversion problem in real time systems? How this problem can be solved? [5]
b) Explain why PCI/X buses are used for high speed data transfer. [5]
c) How to represent Precedence constraints and data dependency among real-time tasks? Explain with diagram. [6]

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OR

- Q6)** a) Explain message queues with suitable diagram. [5]
b) Explain Serial protocols RS-232C & RS-485. [5]
c) Explain in detail Parallel ports I/O Interfacing with neat diagram. [6]

- Q7)** a) What are various Real-time requirements in the domain of Signal processing or Multimedia. [6]
b) What are the different types of semaphores and where they are used? [5]
c) Write Short notes on RT Linux, Vx Works? [5]

OR

- Q8)** a) What is Semaphore? How does it help in resource sharing in RTOS Kernel? [6]
b) Explain semaphores message queues, mailboxes? [5]
c) How interrupts are handled in RTOS environment? [5]

- Q9)** a) Explain with example Validation and debugging in an embedded system. [6]
b) Draw and explain model of real-time communication with related terminologies. [6]
c) List capabilities of commercial real-time operating systems. Enlist the features of RTLinux. [6]

OR

- Q10)** a) What are issues in resource reservation. [6]
b) Explain Resource reservation protocol with diagram. [6]
c) Describe the embedded software development process. [6]

